

Kdb-D2CFR: Solving Multiplayer imperfect-information games with knowledge distillation-based DeepCFR

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ABSTRACT

Counterfactual regret minimization (CFR) is a popular method for finding approximate Nash equilibrium in imperfect-information games (IIG). However, CFR based methods for the IIG are either only designed for two-player IIGs or require much expert knowledge. In this paper, towards to solve the multiplayer IIG problem without utilizing much expert knowledge, we proposed a practical knowledge distillation based framework, which aims to transferring the knowledge from model of two-player IIG into the multiplayer one. By this framework, both of the training efficiency and performance is improved. To eliminate the requirement of expert knowledge, here we introduced a deep learning based CFR in the framework, by which the counterfactual value of CFR can be estimated in an end-to-end way without any expert knowledge and abstraction. We further propose kdb-DeepCFR and kdb-D2CFR based on DeepCFR and D2CFR respectively, which can effectively solve the strategy of multiplayer large-scale game problems. The extensive experiments conducted on 3-8 players poker games suggest that our method outperforms other baselines in the game performance.

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1. Introduction

As a research domain of artificial intelligence, computer game mainly studies the theory of intelligent decision-making [1–3]. Games have been regarded as the touchstone to test the development of artificial intelligence. Generally, compared with perfect-information game (PIG), imperfect-information game (IIG) is more complex because it contains private information unobservable to other players. For example, poker game has already served as a popular benchmark of the IIG [4–12], since it contains all necessary elements of the IIG.

For the two-player IIG, a typical solution is to find its Nash equilibrium or approximate Nash equilibrium. A Nash equilibrium is a list of strategies where no player can improve by deviating to a different strategy [13]. The counterfactual regret minimization (CFR) [14] and its variant methods [11,15,16], which aim to find such strategies with approximate Nash equilibrium, have already achieved great success in the field of IIG. However,

CFR has to traverse the whole game tree, which leads to high computational and storage complexity. Such problem limits its further application in large-scale IIGs. The traditional solution is to abstract the original large-scale game into a small-scale game, then solving the strategy of the small-scale game with CFR. In this way, although the large-scale IIG can be solved with CFR, this solution requires much expert knowledge and causes certain information loss. A recent method DeepCFR greatly reduces the requirement of expert knowledge by applying deep neural networks to CFR [17]. By training the DeepCFR in an end-to-end way, IIGs with two-player can be solved directly without any abstraction. Double Neural CFR (DNCFR) [18], single DeepCFR (SD-CFR) [19] and deep dueling CFR (D2CFR) [20] are all deep learning [21–23] based methods similar to the DeepCFR in recent years.

Although there have been many successful studies in IIGs, it should be noted that they only focus on two-player games [24–28]. For the IIG with multiplayers (eg. Poker games with eight players), which is usually much more complex than two-player IIGs, how to solve it effectively is still a problem worthy of study. Although deep reinforcement learning based methods have achieved successes in some games like StarCraft, Dota and Honor

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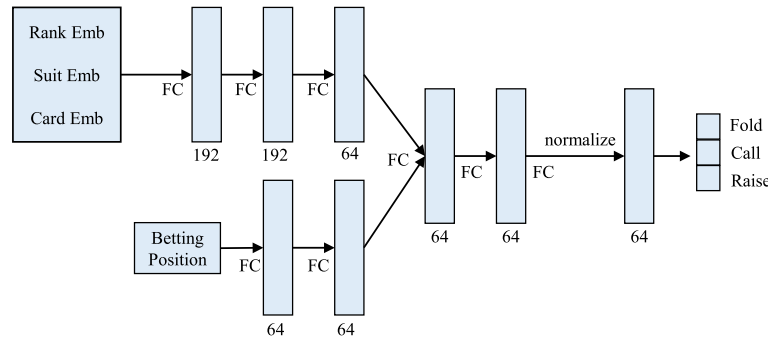


Fig. 1. The neural network architecture of DeepCFR. The network takes an infoset (observed cards and bet history) as input and outputs values (advantages or probability logits) for each possible action. The cards are represented as the sum of three embeddings: a rank embedding (1–13), a suit embedding (1–4), and a card embedding (1–52). Each betting position is encoded by a binary value specifying whether a bet has occurred, and a float value specifying the bet size.

of Kings [29–31]. However, its network training needs large-scale computing support, and its model is not interpretable. In addition, reinforcement learning, especially deep reinforcement learning, requires complex super parameter tuning during training, such as learning rate, discount factor, cache size, etc; Moreover, its convergence is not only lack of theoretical guarantee, but also the training of the network needs extra computing power. For example, AlphaStar [31] needs 16 TPUs for data sampling and network training. In the research on poker games, CFR based methods are the most popular and dominant, with unprecedented success, such as DeepStack [15], Libratus [16] and Pluribus [32]. Pluribus [32] is first superhuman agent in six-player no-limit Texas hold'em game, which solves the game through the combination of abstraction technique and CFR methods. It only focuses on solving six-player games and it needs to be redesigned when the number of player changes. Due to the fact that finding a Nash equilibrium in multiplayer IIGs is at least very hard and even approximating a Nash equilibrium is hard [33]. Thus the goal of solving multiplayer IIGs changes from finding a Nash equilibrium to improving its game performance as much as possible. Despite the game performance of Pluribus surpasses that of professional players, it requires much expert knowledge for detailed design and abstraction techniques, which is a great deficiency for further research.

In this paper, we study the problem of IIG with variant number of players without any expert knowledge and abstraction technique. Following with Pluribus, here we also aim to improve the game performance of the solving method, instead of approximating a Nash equilibrium with theoretical guarantee. To solve the cold-start training problem of vary number of players in IIG, a method of reusing the knowledge of two-player game to multiplayer is proposed. Here the motivation is that in many cases, although the number of players changes, much knowledge in two-player games can be utilized to warm up the training for multiplayer. Thus we propose a knowledge distillation based multiplayer IIG method that can learn the strategy from a two-player IIG model. We summarize our contributions as follows:

1. We design a framework studying on solving the strategy of multiplayer IIGs, which combines CFR based methods and knowledge distillation. The framework can make the knowledge in two-player IIGs shift to multiplayer IIGs by knowledge distillation.
2. We present kdb-DeepCFR and kdb-D2CFR, which can reduce the requirement of expert knowledge in traditional CFR methods. To the best of our knowledge, the kdb-DeepCFR and kdb-D2CFR are the first DeepCFR based method used to solve multiplayer IIGs.
3. Extensive experimental results show that the proposed method performs better than other baselines in both of the two-player and multiplayer IIGs.

The rest of our paper is organized as follows. In Section 2, concepts of extensive-form game, DeepCFR and knowledge distillation are introduced. Our proposed method is described in Section 3, which includes details of the kdb-DeepCFR and kdb-D2CFR. In Section 4, we evaluate the performance of the kdb-DeepCFR and kdb-D2CFR. Finally, in Section 5, we present the conclusions of the paper.

2. Background

2.1. Extensive-form game

An extensive-form game is a model of game that describes sequential-decisions in the field of IIGs. Formally, a finite extensive-form game contains six components $\langle N, H, P, f_c, I, u_i \rangle$ [34]: $\langle N, H, P, f_c, I, u_i \rangle$: player i represents a finite set N of game players, $N = \{1, 2, \dots, n\}$. A node (i.e., history) h is defined by all information of the current situation, including private knowledge known to only one player. A finite set H of sequences, the possible histories of actions, such that the empty sequence is in H and every prefix of a sequence in H is also in H . $A(h) = \{a \mid (h, a) \in H\}$ are actions available after a nonterminal history $h \in H$. $Z \subseteq H$ are terminal histories. P is the player function. $P(h)$ is the player taking action a after history h . $P(h) = c$ represents that the chance determines the action after history h . A function f_c that associates with every history h for which $P(h) = c$ a probability measure $f_c(\cdot \mid h)$ on $A(h)$. The set $I_i \in \mathcal{I}_i$ is an information set of player i . For any information set I_i , all nodes $h, h' \in I_i$ are indistinguishable to player i . Payoff function u_i defines the payoff of terminal state z for each player i . For a zero-sum game, there is $u_1 + u_2 = 0$.

In an extensive-form game, a strategy $\sigma_i(I)$ of player i is a probability vector over actions on information set I . A set of strategies for players, $\sigma_1, \sigma_2, \dots, \sigma_n$, makes up a strategy profile σ , and σ_{-i} represents the strategy in σ except the strategy σ_i of player i . In addition, $\pi^\sigma(h)$ is the probability with h occurs if all players play according to the strategy σ , and $\pi^\sigma(I) = \sum_{h \in I} \pi^\sigma(h)$. $\pi_i^\sigma(h)$ is the contribution of player i to this probability. And formally, $\pi_i^\sigma(h) = \prod_{i \in N \cup \{c\}} \pi_i^\sigma(h)$. Accordingly, $\pi_{-i}^\sigma(h)$ of history h is the contribution of all players (including chance player) except player i .

2.2. DeepCFR

DeepCFR [17] is an improved variant of CFR, which obviates the need for abstraction by instead using deep neural networks to approximate the behavior of CFR in the full game. The neural network architecture of DeepCFR is shown in Fig. 1. DeepCFR computes an approximation of Nash equilibrium by applying the following two steps:

Step 1: Calculate the advantage value. Compared with traditional CFR-based methods, the value network is used to fit the advantage value of player i on each iteration. The advantage value $D_i^T(I, a)$ in DeepCFR is defined as:

$$D_i^T(I, a) = \frac{R_i^T(I, a)}{\sum_{t=1}^T (t\pi_{-i}^{\sigma^t}(I))} \quad (1)$$

where t is the number of iteration, T is the total number of iteration. i is the player and $-i$ is the opponents except the player i . σ^t is the strategy on iteration t . $R_i^T(I, a)$ is cumulative regret of action a on information set I . In DeepCFR, $R_i^T(I, a) = \sum_{t=1}^T (tr_i^t(I, a))$, $r_i^t(I, a)$ is the instant regret of player i for action a on information set I . For instant regret, $r_i^t(I, a) = v^{\sigma^t}(I, a) - v^{\sigma^t}(I)$, and counterfactual value $v^{\sigma^t}(I)$ of player i at information set I is the expected payoff to player i when reaching I , weighted by the probability that player i would reached information set I if she tried to do so that iteration t . Counterfactual action value $v^{\sigma^t}(I, a)$ is the same except it assumes that player i plays action a at information set I with 100% probability.

Step 2: Update the strategy of next iteration. CFR-based methods are iterative method and DeepCFR is no exception. In DeepCFR, regret matching algorithm [35] is used to calculate the strategy of next iteration $t + 1$ and can be described as:

$$\sigma_i^{t+1}(I, a) = \begin{cases} \frac{D_i^{t,+}(I, a)}{\sum_{a' \in A(I)} D_i^{t,+}(I, a')}, & \sum_{a \in A(I)} D_i^{t,+}(I, a) > 0 \\ \frac{1}{|A(I)|}, & \text{otherwise} \end{cases} \quad (2)$$

where $D_i^{t,+}(I, a) = \max(D_i^t(I, a), 0)$.

In DeepCFR, the training data of $D(I, a)$ comes from the external sampling MCCFR (ES-MCCFR) [36], which traverses the game tree and produces the instant regret values. All training data are stored in a memory buffer B_i^v , which is updated through reservoir sampling [37] when its maximum capacity is reached. In addition, DeepCFR fits a policy network to approximate the average strategy $\bar{\sigma}_i^T(I, a) = \frac{\sum_{t=1}^T (t\pi_i^{\sigma^t}(I)\sigma^t(I, a))}{\sum_{t=1}^T (t\pi_i^{\sigma^t}(I))}$, when the total iteration T is reached.

2.3. Knowledge distillation

The concept of knowledge distillation is put forward by Hinton in 2015 [38], which is to realize the knowledge transfer from complex model to simplified model. The traditional knowledge distillation model “teacher-student” is mainly composed of teacher network and student network, as shown in Fig. 2. The model of “teacher-student” is mainly divided into two steps: first, train a teacher model with strong performance. This model is generally complex, which can be a large and deep network model or a model after the integration of multiple models; Then, under the guidance of the teacher model, train the simplified student model, and introduce the soft target related to the teacher model as a part of the overall loss function, so as to induce the student model to train and learn, and finally realize the knowledge transfer from the teacher model to the student model.

3. Our method

In this section, we first describe the overview of the proposed framework. Then, the detail of proposed kdb-DeepCFR and kdb-D2CFR will be introduced.

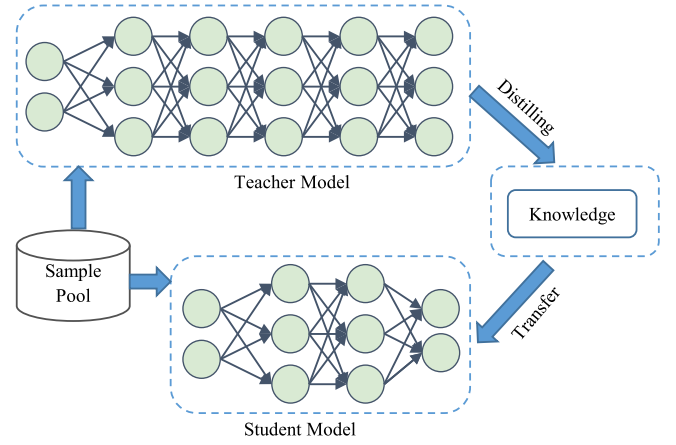


Fig. 2. The traditional architecture of knowledge distilling “Teacher-Student”.

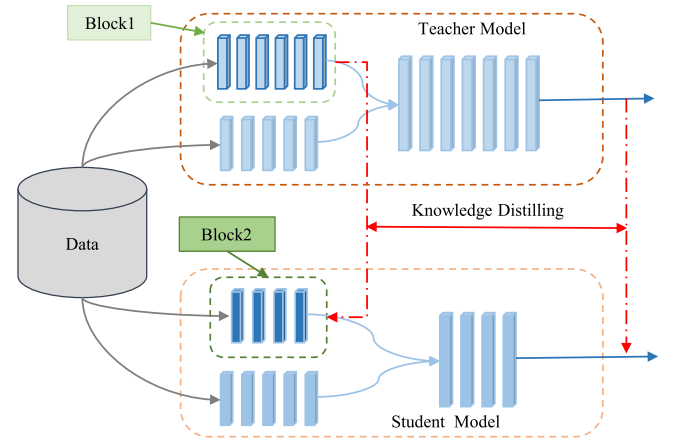


Fig. 3. The framework of Kdb-DeepCFR. The Kdb-DeepCFR is composed of two models: Teacher model and Student model. The two models are both based on the model of DeepCFR.

3.1. An overview of the proposed framework

The above mentioned DeepCFR in Section 2 is proposed only for two-player IIGs. Here we discuss how to adopt the DeepCFR into the IIG with multiplayer. Considering that when dealing with the same type of games, such as poker games, for two-player games and multiplayer games, even if the number of players has increased, there are some common knowledge in the IIGs with different scale. For example, one player with the hand ‘AA’ has a high probability of choosing to raise, whether in two-player or multiplayer poker games. In other words, we can utilize the knowledge of two-players IIG to help training the strategy of multiplayer IIGs. Specifically, in our method, we use knowledge distillation to transfer the knowledge of two-player game to multiplayer game so as to warm up the training of multiplayer game.

In this paper, we propose a framework, knowledge distilling-based DeepCFR (Kdb-DeepCFR), which can transfer the knowledge of DeepCFR in two-player IIGs to solving multiplayer IIGs. The framework of Kdb-DeepCFR is shown in Fig. 3, which is composed of two models: Teacher model and Student model. The two models are both based on the model of DeepCFR. Slightly different from the DeepCFR, we divided its network in the first few layers, which is mainly to facilitate the Block1 of the Teacher model to guide the Block2 of the Student model. It should be noted that, different from the traditional “Teacher-student” knowledge

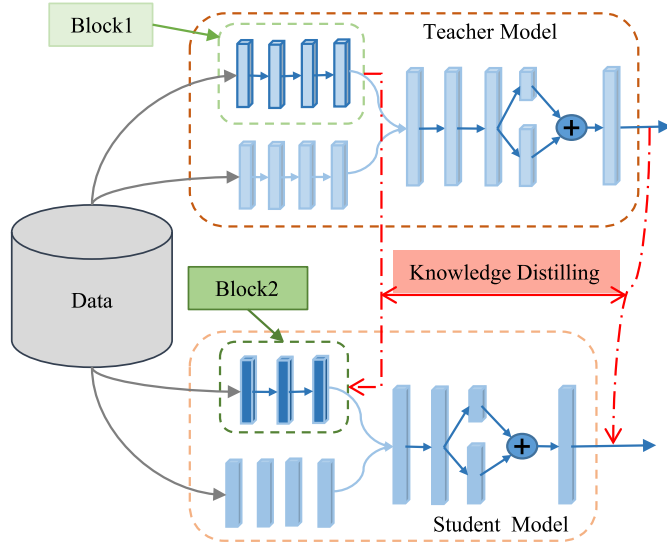


Fig. 4. The framework of Kdb-D2CFR. The Kdb-D2CFR is also composed of two models: Teacher model and Student model. The two models are both based on the model of D2CFR.

distillation methods, the knowledge in the teacher model is transferred from two parts to the student model: from the feature block Block1 to Block2 and from the output of the teacher model to the student model respectively.

Furthermore, similar to the proposed kdb-DeepCFR, we further propose knowledge distilling-based D2CFR (Kdb-D2CFR), which can transfer the knowledge of D2CFR in two-player IIGs to solving multiplayer IIGs. The framework of Kdb-D2CFR is shown in Fig. 4, which is also composed of two models: Teacher model and Student model. The two models are both based on the model of D2CFR, which is the same with kdb-DeepCFR.

Two methods are proposed in this paper, both of which are dedicated to solving multi-player strategy in the field of IIGs. Firstly, they are both strategy solving methods using neural networks combined with CFR, which are a kind of end-to-end strategy solving method from state input to strategy output. Secondly, their solution frameworks are designed based on traditional knowledge distillation, which add feature layer distillation to the classical teacher-student distillation model. Thirdly, the loss functions of their student models are also similar. Finally, they can effectively solve strategies with 3–8 game players in the field of IIGs. Of course, while they have the above similarities, they also have some differences. Firstly, the training methods of value networks in their respective teacher models are different. Secondly, the network structure of the value network in their teacher models is also different. Kdb-DeepCFR uses a fully connected network, while Kdb-D2CFR uses a decoupled network. Finally, their teacher models have different focus. Kdb-DeepCFR is directly fitting regret values through a fully connected network, while Kdb-D2CFR is fitting regret values by focusing on states with higher value through a decoupled network.

3.2. The detail of kdb-DeepCFR and kdb-D2CFR

We will mainly introduce the sample collection and network training in the training process of kdb-DeepCFR. First of all, for the “teacher” model in kdb-DeepCFR, the samples required for training are collected by using DeepCFR in the iterative solution of the two-player imperfect-information games.

Specifically, firstly, in each iteration process, ES-MCCFR is used to traverse the game tree of the specific game. The information set, action regret value samples generated in the traversal

process are stored in the value network sample pool. Secondly, update the iteration strategy of the next iteration through the RM. Thirdly, expand the game tree according to the updating strategy in the next iteration. Store the information set, action samples generated in the iteration process into the strategy sample pool. Considering that DeepCFR needs to approach the value network and strategy network respectively during training, the sample collection here also collects the value samples and strategy samples respectively. When the stored samples reach the upper limit of the sample pool capacity, the reservoir sampling technique is used to update the sample pool.

For the “student” model in kdb-DeepCFR, its samples are collected during the strategy solving process of multiplayer games. Similar to the sample collection process required by the “teacher” model, the only difference is that the “teacher” model collects samples in the process of solving the two-player game strategy, while the “student” model collects samples in the process of solving the multiplayer game strategy.

Training the kdb-DeepCFR: Since the two proposed methods kdb-DeepCFR and kdb-D2CFR are basically similar, we will mainly introduce the training of kdb-DeepCFR here. The network training of Kdb-DeepCFR is carried out from the teacher model and the student model. For the teacher model, its network training can be divided into the training of value network and strategy network respectively. For the value network of the teacher model, its loss function $\mathcal{L}(\theta_i)$ is the MSE (mean squared error) [39] between the output of value network and the samples through traversing the game tree with ES-MCCFR, which can be depicted as:

$$\mathcal{L}(\theta_i) = \mathbb{E}_{(I, t', \tilde{r}') \sim \mathcal{M}_{V,i}} \left[t' \sum_a \left(\tilde{r}'(I, a) - V(I, a | \theta_i) \right)^2 \right] \quad (3)$$

where t' is the number of iteration, $\tilde{r}'(I, a)$ is the regret value of action a on the information set I at t' iteration, $V(I, a | \theta_i)$ is the approximate value of value network for the regret value of action a on the information set I , $\mathcal{M}_{V,i}$ is the sample data of value network.

For the policy network of the teacher model, its loss function $\mathcal{L}(\theta_\pi)$ is the MSE between the output of the policy network and the actual iteration policy, which can be depicted as:

$$\mathcal{L}(\theta_\pi) = \mathbb{E}_{(I, t', \sigma') \sim \mathcal{M}_\pi} \left[t' \sum_a \left(\sigma'(I, a) - \pi(I, a | \theta_\pi) \right)^2 \right] \quad (4)$$

where $\pi(I, a | \theta_\pi)$ is the probability of action a on the information set I for the policy network, $\sigma'(I, a)$ is the probability of action a on the information set I from the iteration strategy $\sigma^{t'}$, $\sigma^{t'}$ is the sample data of policy network.

For the student model, the training process is similar with teacher model except that the guidance of the teacher model is added to the loss function. Specifically, for the value network of the student model, its loss function $\mathcal{L}_{total}(\theta_i)$ can be depicted as:

$$\mathcal{L}_{total}(\theta_i) = \mathcal{L}(\theta_i) + \lambda \mathcal{L}_{block}(\theta_i) + \gamma \mathcal{L}_{output}(\theta_i) \quad (5)$$

where $\mathcal{L}_{total}(\theta_i)$ is the total loss function of the value network, $\mathcal{L}(\theta_i)$ is the hard loss function between the student model and the sample label. $\mathcal{L}_{block}(\theta_i)$ is the loss function between the output of Block1 and Block2. $\mathcal{L}_{output}(\theta_i)$ is the loss function between the output of the teacher model and the student model. λ and γ are hyperparameters.

The loss function $\mathcal{L}_{block}(\theta_i)$ between the output of Block1 and Block2 can be depicted as:

$$\mathcal{L}_{block}(\theta_i) = \mathbb{E}_{(I, t', \tilde{r}') \sim \mathcal{M}_{V,i}} \left[(W_{block1}(x; \theta_{Te}) - W_{block2}(x; \theta_{Stu}))^2 \right] \quad (6)$$

where $W_{block1}(x; \theta_{Te})$ is the output of the Block1 with the parameter θ_{Te} , $W_{block2}(x; \theta_{Stu})$ is the output of the Block2 with the parameter θ_{Stu} .

The loss function between the output of the teacher model and the student model $\mathcal{L}_{output}(\theta_i)$ can be depicted as:

$$\mathcal{L}_{output}(\theta_i) = \mathcal{H}(P_{Te}, P_{Stu}) \quad (7)$$

where $\mathcal{H}$ is cross entropy, $P_{Te}^T = \text{softmax}(\frac{a_{Te}}{T})$ and $P_{Stu}^T = \text{softmax}(\frac{a_{Stu}}{T})$ are the probabilities output by softmax from the value network of the teacher model and student model. a_{Te} and a_{Stu} are the output of the teacher model and student model, T is the temperature coefficient.

For the policy network of the student model, similar with the loss function of the value network, the loss function $\mathcal{L}_{total}(\theta_{\pi})$ can be depicted as:

$$\mathcal{L}_{total}(\theta_{\pi}) = \mathcal{L}(\theta_{\pi}) + \lambda \mathcal{L}_{block}(\theta_{\pi}) + \gamma \mathcal{L}_{output}(\theta_{\pi}) \quad (8)$$

The loss function $\mathcal{L}_{block}(\theta_{\pi})$ of the policy network between the output of Block1 and Block2 can be depicted as:

$$\mathcal{L}_{block}(\theta_{\pi}) = \mathbb{E}_{(I, t', \tilde{r}') \sim \mathcal{M}_{V,i}} [(W_{block1}(x; \theta_{Te}) - W_{block2}(x; \theta_{Stu}))^2] \quad (9)$$

The loss function of the polict network between the output of the teacher model and the student model $\mathcal{L}_{output}(\theta_{\pi})$ can be depicted as:

$$\mathcal{L}_{output}(\theta_{\pi}) = \mathcal{H}(P_{Te}, P_{Stu}) \quad (10)$$

where all parameters are similar to those in the value network settings above.

Training the kdb-D2CFR: The above is mainly an introduction to the kdb-DeepCFR for solving the game strategy of multiplayer imperfect-information games based on knowledge distillation. Considering that kdb-DeepCFR has been introduced in detail from two aspects: sample collection and network training. And the general framework of kdb-D2CFR and kdb-DeepCFR is basically similar, here we only give an overview of this method.

In terms of sample collection, kdb-D2CFR is exactly the same as that in the kdb-DeepCFR. It also collects value network samples by partially traversing the game tree through ES-MCCFR at each iteration. Then update the iterative strategy through the RM, collect the strategy network samples and store them in the sample pool. In terms of network training, although there are differences in network structure between D2CFR and DeepCFR, kdb-D2CFR still separates the value network in the early feature layer according to the framework of kdb-DeepCFR. And distills the knowledge of the early feature layer and the later output layer through knowledge distillation, so as to achieve the knowledge transfer from the “teacher” model to the “student” model. In addition, the network training and model testing of kdb-D2CFR are no different from that of kdb-DeepCFR, thus we do not carry out it in detail in the paper.

4. Experiments

In this section, the experimental setup and experimental results are introduced. The testbed, implementation and evaluation metric will be detailed described in the experimental setup. The comparison experiments and ablation studies are conducted in the experimental results.

4.1. Experimental setup

Experimental testbed: Poker is a family of games that includes hidden information, which has been regarded as the most classic game benchmarks in IIGs in recent years. Many successful CFR-based methods and applications take poker games as the testbed to verify their effectiveness [40–44], such as DeepStack [15], Libratus [16], Pluribus [32].

In this paper, the Heads-up No-Limit Texas hold'em (HUNH) and multiplayer No-Limit Texas hold'em (M-NH) are used to test the effectiveness of the proposed method. The HUNH contains 52 cards in total and consists of four rounds of betting. Each player has two private hands in the first round and totally five public cards in next three rounds. The four betting rounds are preflop, flop, turn and river. Three kinds of actions, fold, call and raise can be chosen by each player on a round of betting. M-NH are exactly the same with HUNH except for the number of game players.

Implementation detail: All experiments in this paper are conducted on the platform OpenSpiel [45], which is a collection of environments and algorithms for research in computer games. The comparison algorithm DeepCFR is trained completely according to the algorithm provided by the OpenSpiel. All parameters are set completely according to the original paper.

We set hyperparameters as follow. For the value network, the information sets are taken as input and outputs the regret value of each action. The policy network outputs probability of each legal actions. The batch size is 200. The parameters are updated by Adam optimizer [46] with a learning rate 0.001. The memory capacity is 100,000. For the Kdb-DeepCFR, the number of iteration is 500, which is enough to verify the effectiveness of the method. In addition, all experiments are conducted on four Xeon(R) CPUs of E5-2640 with 10 cores @2.40 GHz, and one Tesla P100 GPU with 16G memory.

Evaluation metric: In this paper, the effectiveness of our method will be evaluated with a popular metric in the field of IIGs: game performance, just like the experimental setup in previous works [15,16,45].

4.2. Experimental results

In this section, comparison experiments and ablation studies are conducted. For comparison experiments, Head-up no-limit Texas hold'em poker and multiplayer no-limit Texas hold'em poker are used to test the effectiveness Kdb-DeepCFR and Kdb-D2CFR respectively. For ablation studies, the component of Kdb-DeepCFR and Kdb-D2CFR are ablated respectively.

4.2.1. Comparison experiments

The experiment is to verify the effectiveness of Kdb-DeepCFR and Kdb-D2CFR on the M-NH. Specifically, this group experiment is conducted on the number of player from 3 to 8 respectively. For convenience, three-player no-limit Texas hold'em poker is represented by 3-NH in the following, other multiplayer no-limit Texas hold'em poker games are represented in a similar form.

There are few multiplayer baselines available at present in poker games, and since many technical details are lost in their paper (eg. Pluribus [32]), even re-implement some baselines is vary difficult. In this paper four comparison methods the DeepCFR [17], D2CFR [20], DREAM [47] and the agent Jaysen [48] are used in the comparison experiment. It should be noted that the agent here we extend DeepCFR to multiplayer game and Jaysen is the **3rd prize** in 2018-ACPC (Annual Computer Poker Competition, www.computerpokercompetition.org), which is only used on the 6-NH [48] and based on the EHS (effective hand strength) technique [4]. The number of test game is 10,000. The game position will be randomly initialized each time. The game mode adopts one proposed algorithm and multiple same comparison algorithms to match. The Y-axis is the cumulative winning of the proposed algorithm. The positive cumulative winning of the proposed algorithm shows that its game performance is better than the comparison algorithm. The experimental results are shown in Figs. 5–7.

The experimental results on the M-NH of Kdb-DeepCFR vs. DeepCFR and Kdb-DeepCFR vs. DREAM are shown in Figs. 5(a)

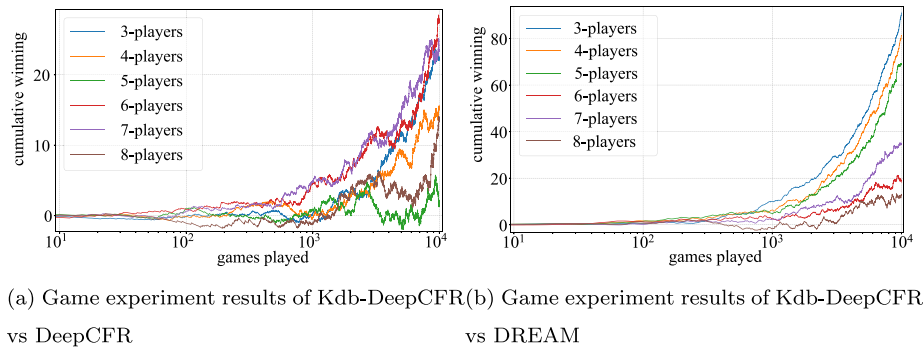


Fig. 5. Game experiment results of Kdb-DeepCFR. The X-axis represents the number of played game, the Y-axis represents the corresponding cumulative winning. The higher the curve, the better the method. The six curves in the figure represents the results tested on various number of game players. For example, “3-players” in left subfigure represents the results of Kdb-DeepCFR vs. DeepCFR on the three-player test game.

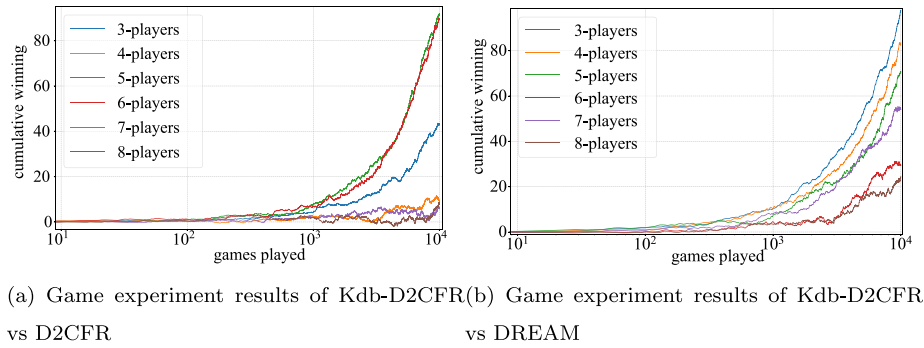


Fig. 6. Game experiment results of Kdb-D2CFR. The X-axis represents the number of played game, the Y-axis represents the corresponding cumulative winning. The higher the curve, the better the method.

and 5(b) respectively. From the curve of cumulative winning, the curves generally show an upward trend with the increase of the number of games. In 10,000 games, the corresponding cumulative winning is positive, and most of the cumulative winning is positive. However, it should be noted that for the 5-NH in Fig. 5(a), the curve of 5-players shows a downward trend in the game revenue at 1500 ~ 1800 games and 800 ~ 3800 games, and an overall upward trend after 4000 games. The curve of 8-players shows that the cumulative winning also has a downward trend at 3500 ~ 6500 games. In Fig. 5(b), it can be found that there is a downward trend at 450 ~ 1200 games in the 8-players. In addition, the curve corresponding to the 7-NH also fluctuates greatly, and the final cumulative winning is relatively small. Nevertheless, from the curve in Fig. 5, it can be found that the game performance of Kdb-DeepCFR is better than the comparison algorithm in the multiplayer game.

The experimental results on the M-NH of Kdb-D2CFR vs. D2CFR and Kdb-D2CFR vs. DREAM are shown in Figs. 6(a) and 6(b) respectively. Similar to the results of Kdb-DeepCFR in Fig. 5, from the curve of cumulative winning, they generally show an upward trend with the increase of the number of games. From Fig. 6(b), we can also find that the cumulative winning shows an increasing trend when playing against the DREAM. In 10,000 games, the corresponding cumulative winning is positive, and most of the cumulative winning in the game is positive. However, it should be noted that from Fig. 6(a), it can be found that the cumulative winning of the 8-players is not only negative at 3500 ~ 5000 games, but also there is a significant gap between the cumulative winning after 10,000 games and that in other multiplayer games. The corresponding curve of the 7-players also fluctuates greatly, and the final cumulative winning is relatively small. Nevertheless, from the curve in Fig. 6, it can be seen that the game performance of kdb-D2CFR is better than the comparison algorithm.

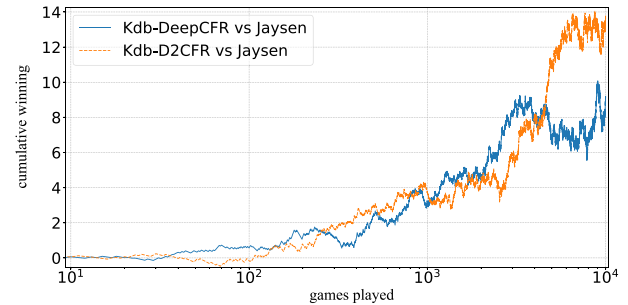


Fig. 7. Game experiment results of Kdb-DeepCFR, Kdb-D2CFR vs. Jaysen on the 6-NH. The X-axis represents the number of played game, the Y-axis represents the corresponding cumulative winning. The higher the curve, the better the method.

Table 1

Game experiment results of Kdb-DeepCFR on the M-NH (mbb/g).

Game player number	3-player	4-player	5-player
vs. DeepCFR	22.1 ± 9.89	17.0 ± 10.99	15.2 ± 12.01
vs. DREAM	39.78 ± 11.48	38.10 ± 11.36	25.43 ± 11.66
Game player number	6-player	7-player	8-player
vs. DeepCFR	27.74 ± 12.83	30.02 ± 13.52	17.33 ± 13.37
vs. DREAM	11.45 ± 8.92	15.83 ± 10.33	10.39 ± 10.52

Tables 1 and 2 respectively record the average winning of the proposed algorithm kdb-DeepCFR, kdb-D2CFR and the comparison algorithm within the 95% confidence interval for 10,000 games. For kdb-DeepCFR shown in Table 1, its winning when compared with the comparative algorithm DeepCFR on three to eight player games are: 22.1 ± 9.89 mbb/g, 17.0 ± 10.99 mbb/g,

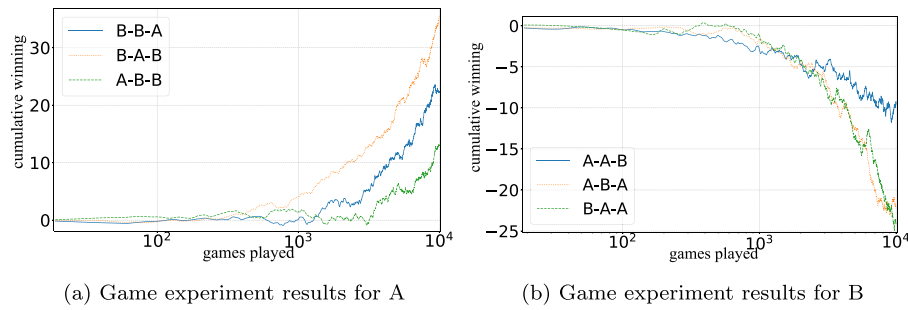


Fig. 8. Test experiment results of position sensitivity. The X-axis represents the number of played game, the Y-axis represents the corresponding cumulative winning. The higher the curve, the better the method.

Table 2

Game experiment results of Kdb-D2CFR on the M-NH (mbb/g).

Game player number	3-player	4-player	5-player
vs. D2CFR	26.02 ± 11.07	15.1 ± 10.66	95.04 ± 14.13
vs. DREAM	39.09 ± 10.36	38.34 ± 12.83	25.89 ± 10.33
Game player number	6-player	7-player	8-player
vs. D2CFR	91.08 ± 16.07	6.71 ± 10.75	10.66 ± 12.94
vs. DREAM	13.93 ± 10.20	19.42 ± 11.64	12.41 ± 9.06

Table 3

Test experiment results of position sensitivity (mbb/g).

For A	B-B-A	B-A-B	A-B-B
Game winning	24.44 ± 9.97	35.30 ± 9.8	13.26 ± 9.58
For B	A-A-B	A-B-A	B-A-A
Game winning	-11.4 ± 9.04	-27.48 ± 9.42	-25.38 ± 10.03

15.2 ± 12.01 mbb/g, 27.74 ± 12.83 mbb/g, 30.02 ± 13.52 mbb/g, 17.33 ± 13.37 mbb/g respectively, and the benefits when compared with the comparative algorithm dream are 39.78 ± 11.48 mbb/g, 38.10 ± 11.36 mbb/g, 25.43 ± 11.66 mbb/g, 11.45 ± 8.92 mbb/g, 15.83 ± 10.33 mbb/g, 10.39 ± 10.52 mbb/g respectively.

For kdb-D2CFR shown in Table 2, its winning when compared with the comparative algorithm D2CFR on three to eight player games are: 26.02 ± 11.07 mbb/g, 15.1 ± 10.66 mbb/g, 95.04 ± 14.13 mbb/g, 91.08 ± 16.07 mbb/g, 6.71 ± 10.75 mbb/g, 10.66 ± 12.94 mbb/g respectively, and the winning when compared with the comparative algorithm DREAM are: 39.09 ± 10.36 mbb/g, 38.34 ± 12.83 mbb/g, 25.89 ± 10.33 mbb/g, 13.93 ± 10.20 mbb/g, 19.42 ± 11.64 mbb/g, 12.41 ± 9.06 mbb/g respectively. It can be found that the proposed kdb-DeepCFR performs better in three player, six player and seven player games than the comparison algorithm. Compared with the comparison algorithm, the proposed kdb-D2CFR has stronger performance in five player and six player games, and relatively poor performance in seven player and eight player games. From the overall data in the Tables 1 and 2, except the poor performance of kdb-D2CFR in seven player and eight person games, the proposed kdb-DeepCFR and kdb-D2CFR have a leading superiority in other multiplayer games.

Fig. 7 shows the cumulative winning results of Kdb-DeepCFR, Kdb-D2CFR against Jaysen on the 6-NH respectively. Among them, the game mode is one Kdb-DeepCFR and five jaysen agents. The initial position is random, and the game mode of Kdb-D2CFR is the same. It can be seen from Fig. 7 that although the cumulative winning of Kdb-D2CFR at 60 ~ 100 games is a negative return, and the cumulative winning of Kdb-DeepCFR has a downward trend at 4000 ~ 6500 games. With the increase of the number of games, its cumulative winning still shows an upward trend. And in most cases, it remains a positive winning. This is enough

to show that compared with Jaysen agent, the proposed Kdb-DeepCFR and Kdb-D2CFR have stronger game performance in the game.

After previous analysis and validation analysis of comparative experiments, in general, compared with previous methods, as far as we know, these two methods are the first to use neural networks combined with CFR to effectively solve multiplayer game strategies, which provides a new idea for the expansion of the CFR-based method in multiplayer research. Secondly, the method we proposed effectively realizes the knowledge reuse from two-player solution to multiplayer solution. And it is also not involved in previous methods related to CFR-based methods, and is a good reference for solving multiplayer with CFR-based methods. In addition, although the proposed method combines a knowledge distillation framework, it is not complex compared to the previous six-player agent Pluribus; Finally, our method can not only effectively solve multiplayer strategies, but also effectively solve game strategies with 3–8 players, compared to previous multiplayer solutions that can only solve a fixed number of game players.

Of course, there are still some areas worth improving in our method. Firstly, the current strategy accuracy of the proposed method needs to be improved. Further improvements to the network architecture or more layers and finer network architecture design may help improve the accuracy of the solution strategy. In addition, the theoretical guarantee of the proposed method is lacking compared to the CFR method. The CFR method can ensure that the solution strategy is an approximate Nash equilibrium strategy in two-player zero-sum games, but it does not have theoretical guarantees in multiplayer games. The same is true of the method we proposed. Finally, the proposed method has only been verified in poker games, especially in Texas Hold'em poker, and its expansion in other types of applications also requires further research.

4.2.2. Ablation studies

This section conducts ablation analysis experiments on the proposed Kdb-DeepCFR and Kdb-D2CFR, mainly from two aspects: firstly, this section analyzes the position sensitivity of the proposed method in multiplayer games, and the experimental results are shown in Fig. 8 and Table 3; Secondly, this section analyzes the effectiveness of knowledge distillation in the proposed method. The experimental results are shown in Figs. 9–12 respectively.

Position sensitivity analysis: In multiplayer games, especially in the specific multiplayer game scenario M-NH in the paper, the final game performance will be affected by the different positions of the game players. In order to conduct an experimental analysis of the proposed method more objectively, this section analyzes the sensitivity of the game players to their positions in the game. The test experiment is carried out on the 3-NH. One Kdb-DeepCFR

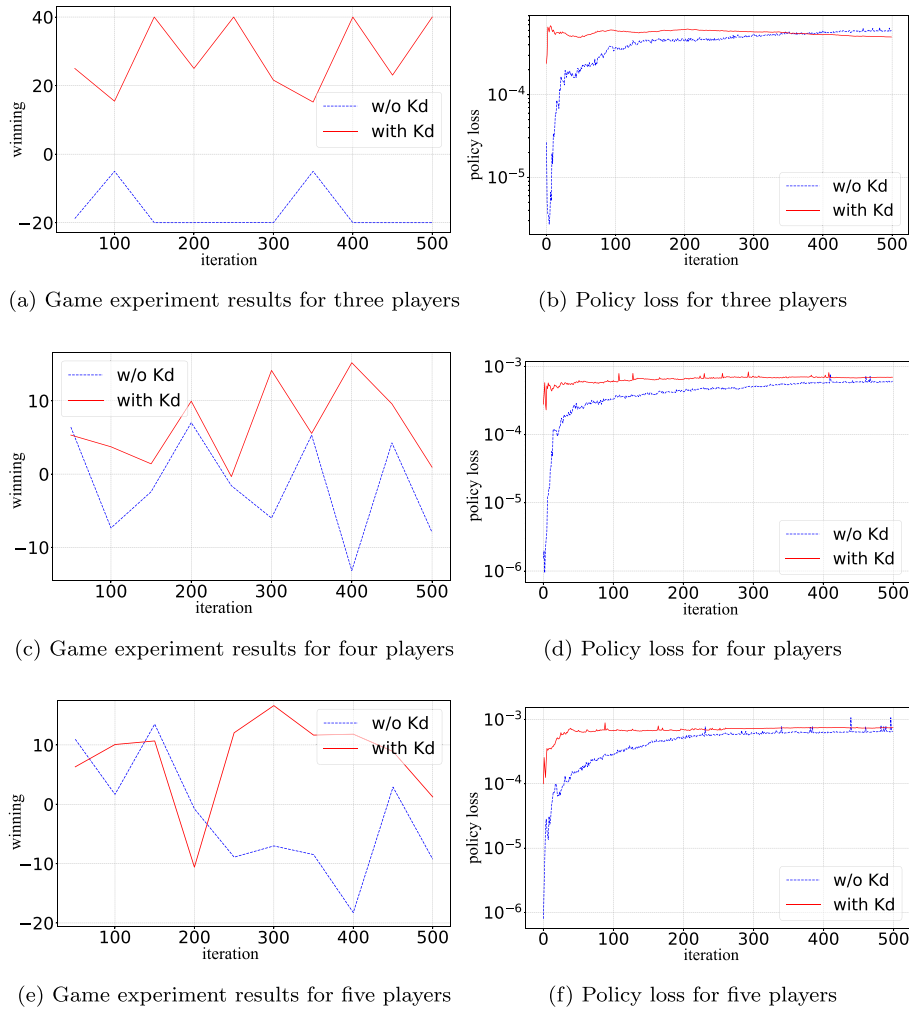


Fig. 9. Ablation experiment results on multiplayer games (3–5) of kdb-DeepCFR.

and two DeepCFR, one DeepCFR and two Kdb-DeepCFR are used for the test game. 10,000 games at each initial position are conducted respectively, and the experimental results are shown in Fig. 8, where the Y-axis represents the cumulative winning of games, and the X-axis represents the number of played games. The higher the curve, the better the result. In Fig. 8, “A” refers to the Kdb-DeepCFR, “B” refers to the comparison algorithm DeepCFR. “A-B-B” means that Kdb-DeepCFR is in the first position and the two rivals DeepCFR are in other positions. Accordingly, “B-A-A” means DeepCFR is in the first position, and the two rivals Kdb-DeepCFR are in other positions. In addition, the winning of the test is the winning of the non-repeated game players among the three game players. That is, “A-B-B” is for the winning of “A”, as shown in Fig. 8(a), and “B-A-A” is for the winning of “B”, as shown in Fig. 8(b).

It is obvious from Fig. 8(a) that their winnings are all positive and generally maintain a gradual growth trend. That is, when one Kdb-DeepCFR is used to be against two DeepCFR, Kdb-DeepCFR is dominant. It shows that the performance of Kdb-DeepCFR is better than DeepCFR, which further verifies the effectiveness of the proposed method. However, it should be noted that the winning curves representing Kdb-DeepCFR in different game positions are not the same, and there is a gap between the three curves. This shows that when Kdb-DeepCFR is in different game positions, its game winning is different. That is, different game positions will affect the winning of game players. It can be seen from Fig. 8(a) that when Kdb-DeepCFR is in the middle position, the winning is

the highest, and when it is in the first place, the winning is the lowest.

As for the experimental results of “B-A-A” using one DeepCFR and two Kdb-DeepCFR, as shown in Fig. 8(b). It can be found that the winnings of the three curves are negative, and they are generally in a smaller and smaller trend. This shows that the game performance of DeepCFR is weaker than that of Kdb-DeepCFR. Secondly, although the curve “A-B-A” is basically consistent with the curve “B-A-A”, there is still a significant gap between them. And the curve “A-A-B” further shows that different game positions will affect the corresponding winnings of game players. It can be seen from Fig. 8(b) that when three players play the game, the loss of DeepCFR is the smallest when it is at the end, and the loss is larger when it is at the first and middle positions.

Table 3 further records the relevant data. The table records the average winning of the method within 95% confidence interval after 10,000 games at different positions, in units of mbb/g. It can be seen that when one Kdb-DeepCFR is used to be against two DeepCFR, the average winning is $35.30 \sim 9.8$ mbb/g when the position is “B-A-B”. While the average winning is $24.44 \sim 9.97$ mbb/g and $13.26 \sim 9.58$ mbb/g when the position is “B-B-A” and “A-B-B”, respectively. When the position is “B-A-B”, the winning is the highest, which also corresponds to the curve in Fig. 8(a). When using one DeepCFR and two Kdb-DeepCFR, the winnings at positions “A-B-A” and “B-A-A” are relatively close, which are $-27.48 \sim 9.42$ mbb/g and $-25.38 \sim 10.03$ mbb/g respectively. When the position is “A-A-B”, the winning is $-11.4 \sim 9.04$ mbb/g, which is relatively small compared with the first two.

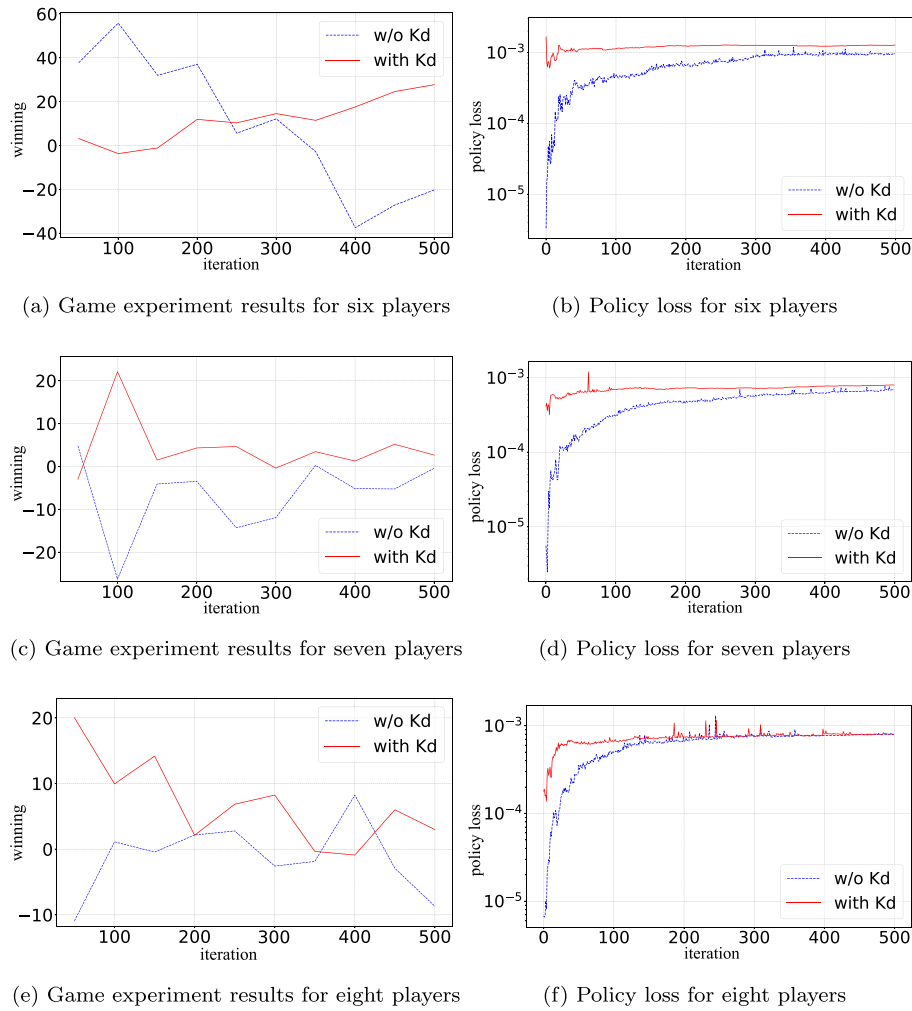


Fig. 10. Ablation experiment results on multiplayer games (6–8) of kdb-DeepCFR.

In general, through the experiment of position sensitivity analysis, the experimental results in Fig. 8 and Table 3 show that different game positions have an impact on the winnings of game players. In order to make the comparison experiment and other related experiments more objective and fair, all other experiments carried out in the paper randomly select the game position of game players, so as to reduce the impact of different game positions on the winning of game players.

Ablation analysis of knowledge distillation: Knowledge distillation is an important part of the proposed method. In this section, ablation analysis experiments are carried out for the knowledge distillation to verify its effectiveness. The policy network loss and the game performance in the iteration process are used as the evaluation metrics. Game performance refers to the average winning of 1000 games with the corresponding policy network every 50 iterations in the training process. In order to fully verify the effectiveness of the knowledge distillation, the experiment was tested on three to eight player games respectively.

In addition, it should be noted that considering the previous position sensitivity analysis, it is found that not only the different positions will affect the game performance, but also the game mode selection of “1 A vs. 2B” and “1B vs. 2A” will also affect the game performance. For this reason, the “1 A vs. 2B” mode is adopted here. That is, the test method is one, and the comparison method is the same method of multiple opponents. The experimental results are shown in Figs. 9–12.

For the game performance, “with Kd” refers to the proposed Kdb-DeepCFR with knowledge distillation, “w/o Kd” refers to Kdb-DeepCFR without knowledge distillation. Their corresponding curves respectively represent the game winnings of playing against Kdb-DeepCFR when the opponents are all DeepCFR and the game winnings of playing against DeepCFR when the opponents are all Kdb-DeepCFR. Therefore, there is no constraint relationship between the two curves here. The curve only reflects the game performance of their respective methods as the only test method, which can better reflect the impact of knowledge distillation on the method under the same conditions. For policy loss, it means that the loss of each network changes with the number of iterations in the corresponding multiplayer game. Figs. 11 and 12 show the relevant ablation experiments of kdb-D2CFR, and all settings are consistent with those in Kdb-DeepCFR.

Figs. 9 and 10 show the experimental results of kdb-DeepCFR after ablation analysis on 3-NH to 5-NH and 6-NH to 8-NH, respectively. For the game performance, as shown in Figs. 9(a), 9(c), 9(e), 10(a), 10(c) and 10(e). It can be found that the “with Kd” represented by the red curve is basically in a positive winning during the game, while the blue dotted line representing “w/o Kd” is in a negative winning in most cases, which shows the effectiveness of the knowledge distillation of the proposed method when playing in the same game mode. However, it should be noted that first of all, the curve fluctuates greatly, which reflects the great randomness of the experimental results. This is because

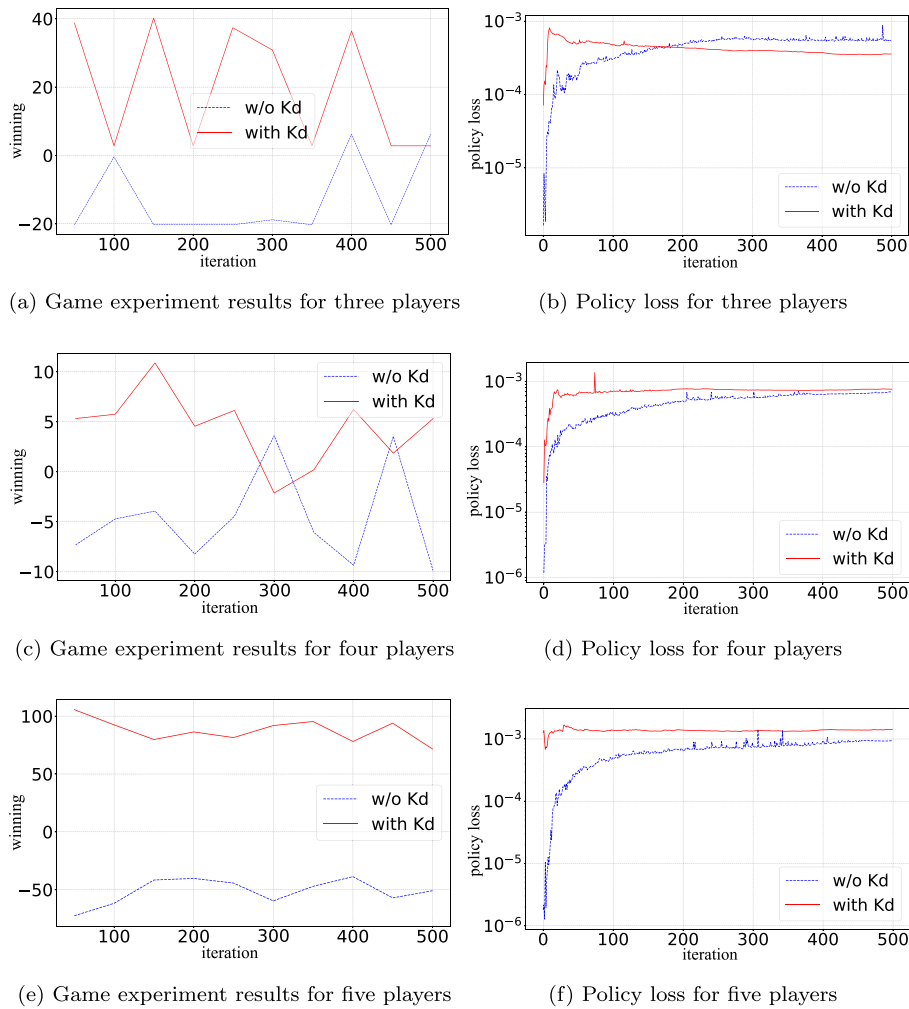


Fig. 11. Ablation experiment results on multiplayer games (3–5) of kdb-D2CFR.

the position of the method is set randomly during the game in the experiment. In the previous position sensitivity analysis experiment, it has been found that different positions have an impact on the game winning. However, in order to make the experimental results more objective, it is still necessary to randomly set the game position.

For the 3-NH, it can be found from Fig. 9(a) that the game conducted under different iterations are all positive winnings; For the 4-NH, it is found from Fig. 9(c) that although its game winning is slightly less than zero at the 250th iteration, it is still in the majority of positive winnings on the whole; For the 5-NH, its game performance is similar to that in the 4-NH. From Fig. 9(e), it is found that it is a negative winning in the 200th iteration and a positive winning in other iterations; For the 6-NH, it is found from Fig. 10(a) that it is still a positive winning except for the negative winning in the 100th and 150th iterations; For the 7-NH, it is found from Fig. 10(c) that the winning is negative at 50th iteration and 300th iteration; For the 8-NH, it is found from Fig. 10(e) that the winning is negative at 350th iteration and 400th iteration.

Figs. 11 and 12 show the experimental results of kdb-D2CFR after ablation analysis on 3-NH to 5-NH and 6-NH to 8-NH, respectively. For the game performance, as shown in Figs. 11(a), 11(c), 11(e), 12(a), 10(c) and 12(e). It means that the red curve of kdb-D2CFR performs well in the three to six player game, their game winning is basically positive and has a great leading advantage. Although the winning in the 300th iteration of the 4-NH is negative, the winning in other iterations is still positive.

In the 7-NH and the 8-NH, the trend of the game winning is similar to that of kdb-DeepCFR in the 7-NH and the 8-NH. On the whole, the game performance in the seven and eight player game is lower than that in the three to six player game.

In general, it is obvious that the knowledge distillation is helpful for the game performance of the method. However, it should also be noted that in the 7-NH and 8-NH, the game winning has been very close to zero, which indicates that the effect of the method in the 7-NH and 8-NH is limited. This is because although the knowledge distillation is effective, it has a good performance in the three to six player game, which shows that the student model does learn the “knowledge” in the teacher model. However, this kind of “knowledge” also has its own limitations, and its effectiveness decreases with the increasing number of game players. Specifically, in this section of the experiment, we can find that the game performance of the proposed method on the 7-NH and 8-NH decreases.

For the policy loss, the results are shown in Figs. 9(b), 9(d), 9(f), 10(b), 10(d) and 10(f). They are the curve of kdb-DeepCFR’s policy network loss with or without knowledge distillation on the three to eight player game with the number of iterations. Figs. 11(b), 11(d), 11(f), 12(b), 10(d) and 12(f) show the corresponding test curves of kdb-D2CFR on the three to eight player game. It can be found that no matter in kdb-DeepCFR or kdb-D2CFR, the loss of the corresponding policy network with/without knowledge distillation tends to stabilize with the increase of the number of iterations. It not only shows that the policy network has converged in

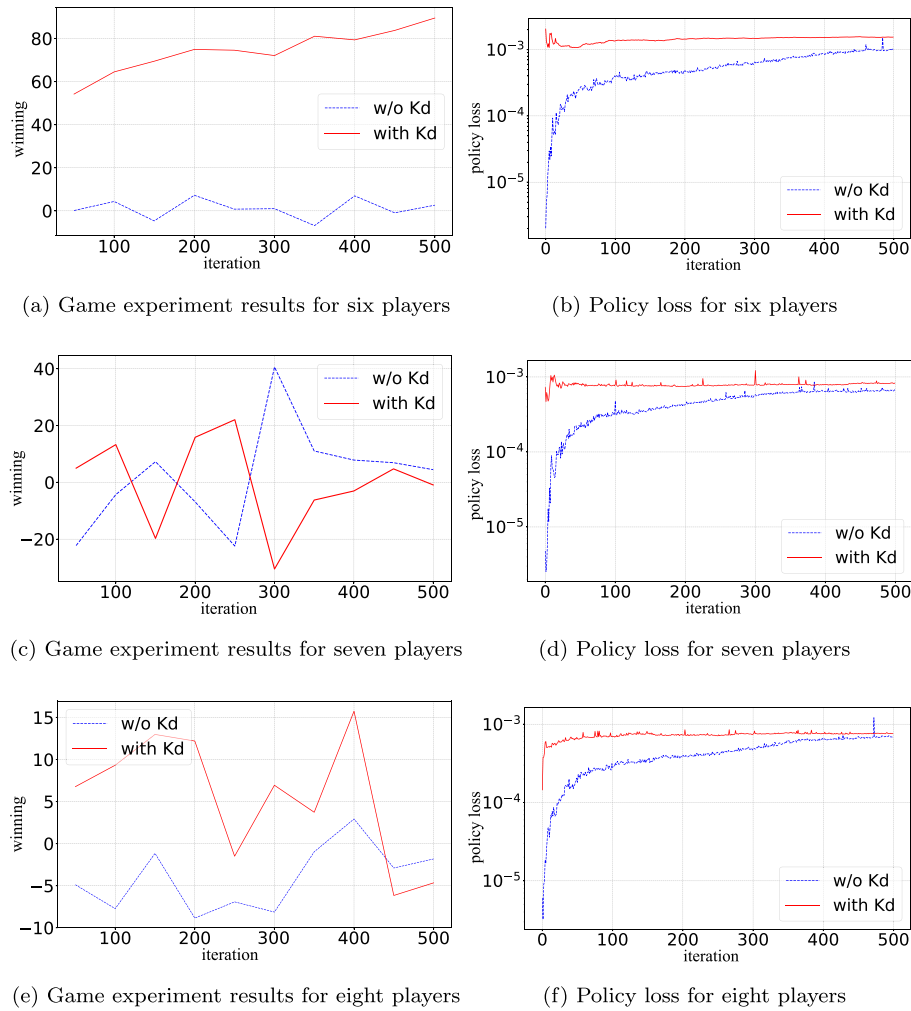


Fig. 12. Ablation experiment results on multiplayer games (6–8) of kdb-D2CFR.

the later stage of the iteration, but also shows that the parameter setting of the number of iterations in this experiment is reasonable. The 500 iterations is enough to reflect the quality of the experimental results in this section. For the red curve “with kd”, it can be found from the figure that compared with the blue curve “w/o kd”, the corresponding policy network has achieved convergence with fewer iterations. In addition to the three player game in Fig. 11(b), kdb-D2CFR converges around the 400th iterations. In other multiplayer games, both kdb-DeepCFR and kdb-D2CFR have converged at most 200th iterations. In most multiplayer games, the policy network has converged within 100th iterations. This fully verifies the effectiveness of the knowledge distillation of the proposed method.

5. Conclusion

In this paper, in order to deal with the problem of IIG with variant number of players without any expert knowledge and abstraction technique, we present kdb-DeepCFR and kdb-D2CFR to solve the strategy of multiplayer IIGs. The methods combine CFR based methods and knowledge distillation, which can make the knowledge in two-player IIGs shift to multiplayer IIGs by knowledge distillation. As far as we know, the kdb-DeepCFR and kdb-D2CFR are the first DeepCFR based method used to solve multiplayer IIGs. The extensive experimental results show that the proposed methods are effective, and outperforms other comparison methods on multiplayer poker games. In the future,

we will further improve the game performance of this method through designing network structure and network parameters. In addition, it is also worth studying to extend this method to other imperfect-information games. Furthermore, it is also a very worthwhile work to explore the convergence theory of equilibrium strategies in multi-player IIGs.

CRedit authorship contribution statement

Huale Li: Conceptualization, Methodology, Data curation, Writing – original draft. **Zengyue Guo:** Data curation, Software. **Yang Liu:** Software, Validation. **Xuan Wang:** Writing – reviewing. **Shuhan Qi:** Writing – review & editing. **Jijia Zhang:** Supervision. **Jing Xiao:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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